

residues in hard-to-reach locations. Each equipment manufacturer should be able to produce a brief written report describing the process, results, fluids, temperatures and times. Disqualify any system that cannot clean successfully.

At this point in your equipment search, the above four criteria should produce a short list of cleaning systems that fit your company's requirements. Now comes the hard part: determining which system will produce clean parts at the lowest total cost.

Three types of costs

Once the throughput has been defined, the engineer must research the direct operating costs, the indirect operating costs and the fixed costs for each alternative. Direct operating costs include the cost of consumables and labour required by the cleaning system when it is operating. Importantly, when the machine stops, so do the costs. Direct costs are directly proportional to throughput.

Indirect costs are those costs that occur even when the machine is not cleaning, such as rent and training.

Fixed costs are usually one-time installation capital costs.

Let's take a look at each of these costs:

Direct operating costs. Electrical costs are a big issue for most cleaning systems. Typically, the total electrical costs of an aqueous system are five to ten times the costs of a vapour degreaser due to the need to heat the water, pump the water, spray the water and dry the water. Waste-water treatment systems are also very energy-hungry.

Water is cheap to buy but expen-



Fig. 3: Vapour degreaser in laboratory

sive to re-purify. Solvents and saponifiers can be expensive and are subject to drag-out. (Drag-out is wasted cleaning fluid that is trapped inside or around the parts as they are removed from the cleaning system.) Vendors should be able to estimate fluid consumption based on the sample parts provided to them.

Big machines have big maintenance problems, and large aqueous systems have the most complexities. Filters must be checked and replaced. Blowers, motors and conveyors must be maintained. Complex water-treatment and recycling processes must be monitored. Alkaline additives boost the cleaning power of many systems; these additives coat the machine's interior and cause additional maintenance problems. All of these should be included in the system's direct operating costs.

Labour costs should be carefully tabulated. Most of the labour is involved in loading and unloading the systems, which is pretty straightforward. Also look for time wasted by technicians in performing auxiliary inspections, such as hand-spraying, re-cleaning or hand-drying products outside the machine. In today's world, manual intervention should be rare. If it is not, something is wrong. To measure hourly labour costs, many companies use fully-loaded labour rate

for the technicians who will operate the system.

Indirect costs. Indirect labour can be difficult to measure. This might include the cost of a supervising engineer's time, the cost of training, the hourly cost of maintenance technicians and the cost of any chemical safety training. If employee turnover is a problem, add additional funds for quarterly supplemental training.

There is an essential element to indirect labour that is often ignored: the complexities of managing water-based cleaning systems require many hours of well-trained, experienced technical experts. Their time is spent managing the water purification systems, the pH and additives of the water in the system, and the resulting waste-water treatment systems. The commitment in engineering required to operate a water system properly typically results in ongoing costs that far surpass those of vapour systems.

Another indirect cost is floor space, and sometimes a large one. Aqueous and semi-aqueous systems require more floor space than solvent-based systems and water-treatment facilities. These systems can be as big as the cleaners themselves. They also have slower cycle times, so more space is needed for storage, work-in-progress,

Cleaning scoreboards

There are many different types of sports scoreboards, but they all summarise the performance of the players and indicate who performed the best on that particular day. Engineers also need a scoreboard, to help compare the different types of cleaning systems.

There are four main types of cleaning systems used in electronics assembly—aqueous systems, semi-aqueous systems, hydrocarbon systems and vapour degreasing systems—and they vary widely in size, performance, operating costs, labour costs and more. A clever cleaning scoreboard helps engineers decipher these complexities and make apple-to-apple comparisons of very different types of machines.